

Total No. of Questions : 6]

SEAT No. :

P4859

[Total No. of Pages : 3

T.E./Insem. - 146
T.E. (Computer Engg.)
THEORY OF COMPUTATION
(2012 Pattern) (Semester -I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer (either (Q 1 or Q 2) and (Q3 or Q4) and (Q5 or Q6)).
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.
- 4) Give suitable examples wherever necessary.

UNIT - I

Q1) a) Describe the languages accepted by the following Regular Expressions and justify with suitable examples: **[6]**

- i) a. $(a+b)^* .ab$
- ii) $(1^*.0.1^*.0.1^*)^*$
- iii) a^*b+b^*a

b) Prove by mathematical induction the following: **[4]**

$$\text{For all } n \geq 1, \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

OR

Q2) a) Apply the theorem of Pumping Lemma to prove that the following language L is regular. **[6]**

$$L = \{ a^{2^n} \mid n \geq 1 \}$$

Draw the finite automata for L.

P.T.O.

- b) Draw FA for strings which do not contains '00' as substring in it over alphabet $\Sigma = \{0,1\}^*$ [2]
- c) Draw NFA for strings ending with '10' over alphabet $\Sigma = \{0,1\}^*$. [2]

UNIT - II

- Q3)** a) Convert the following regular expression to its equivalent DFA: [8]
 $(a + b)^* .abb$
- b) What are the limitations of Finite Automata? Justify with suitable examples. [2]

OR

- Q4)** a) Convert following NFA into its equivalent DFA [4]

$\Sigma - >$	0	1
Q		
$\rightarrow P$	P,Q	R
Q	R	R
R	S	Q
*S	S	S

- b) Obtain the regular expression that denotes the language accepted by the following DFA, using Arden's Theorem: [4]



- c) Give the formal definition for a Moore Machine, with a suitable example. [2]

UNIT - III

Q5) a) Give context free grammars for the following languages: [6]

i) $L = \{ a^n b^{2n} \mid n \geq 0 \}$

ii) $L = \{ 0^i 1^j 0^k \mid j \geq (i + k) \}$

b) Simplify the CFG given below, by eliminating all unit productions: [4]

$$S \rightarrow AB \mid bB$$

$$A \rightarrow a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow D \mid bC \mid a$$

OR

Q6) a) Convert the given right-linear grammar to its equivalent left-linear form:[6]

$$S \rightarrow aA \mid bB$$

$$A \rightarrow bC$$

$$B \rightarrow aC$$

$$C \rightarrow aC \mid bC \mid a \mid b$$

b) Explain the closure properties of Context-free Languages (CFLs). [4]

